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ENVIRONMENT RESEARCH

CASE STUDY

Aeration in sludge holding tanks as an economical means for biosolids odor control—A case study

Hao Luo[1] | **Dian Zhang**[1]* | **Malcolm Taylor**[2]* | **Caroline Nguyen**[2]* | **Zhi-Wu Wang**[1]*

[1]Department of Civil and Environmental Engineering, Center for Applied Water Research and Innovation, Ashburn, VA, USA

[2]Washington Suburban Sanitary Commission, Laurel, MD, USA

Correspondence
Zhi-Wu Wang, Virginia Tech, 44961 Loudoun Water Way, Ashburn, VA 20147, USA.
Email: wzw@vt.edu

Abstract

Abnormally high-odor detection threshold (DT) values were detected for biosolids produced at one of the water resource recovery facilities (WRRFs) of Washington Suburban Sanitary Commission. As an inexpensive countermeasure, aeration of thickened sludge holding tanks (SHTs) was tested as a solution for mitigating the subsequent biosolids odor emission. Experimental results indicated that the extremely low-oxidation reduction potential (ORP) in the SHTs and the fermentation of high-rate-activated sludge were primarily contributors to the odor emission from the de-watered cake. Two rounds of bench-scale experiments on different days confirmed that aerating the sludge in holding tanks reduced peak emission concentrations of sulfurous odorous compounds such as hydrogen sulfide (H_2S), methanethiol (MT), and dimethyl sulfide (DMS) from 203, 110, and 20 mg m^{-3} g^{-1} dry to 119, 70, and 14 mg m^{-3} g^{-1} dry, respectively. Further preliminary full-scale validation study showed that even a slight ORP improvement from −180 mV to −162 mV reduced the peak H_2S concentration from 87 to 48 mg m^{-3} g^{-1} dry and decreased the biosolids DT value from 4266 to 1862. It was concluded that lifting ORP in SHTs through aeration can be used by utilities as a simple means for biosolids odor control.

Practitioner points

- Anaerobic storage of high-rate active sludge was the main reason for the excessive biosolids odor.
- Aeration of sludge holding tanks can effectively reduce biosolids odor.
- A slight oxidation reduction potential improvement substantially reduced biosolids odor.

KEYWORDS

anaerobic, DMS, H_2S, MT, odor detection threshold, ORP

INTRODUCTION

A water resource recovery facility (WRRF) administrated by the Washington Suburban Sanitary Commission (WSSC Water) has been facing the pressing challenge of excessive biosolids odor. The biosolids from this WRRF, which has an average capacity of ~30 million gallons per day, are not currently permitted as Class B biosolids, limiting their disposal options to landfilling. Over the past several years, landfills have expressed reluctance to accept biosolids from this WRRF and rejected loads in certain instances. WSSC Water is therefore actively looking for cost-effective

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strategies to reduce biosolids odor and decrease the risk of further increasing costs associated with solids hauling and disposal. Beginning in 2024, dewatered cake from this and other WSSC Water WRRFs will be hauled to WSSC Water's centralized bioenergy facility to be further treated through thermal hydrolysis and anaerobic digestion. Because of this, WSSC Water elected to focus on solutions that require relatively low-capital investment.

As illustrated in Figure 1, this WRRF has a three-sludge biological process: (1) high-rate activated sludge [HRAS], (2) nitrification activated sludge [NAS], and (3) denitrification activated sludge [DNAS]. Waste sludge streams from these three processes are combined and then thickened using dissolved air flotation (DAF). Following the DAF, thickened sludge is stored in sludge holding tanks (SHTs) before being dewatered with centrifuges which are operated 24 h day^{-1} and 5 days $week^{-1}$. Because the dewatering system is offline during weekends, thickened sludge from the DAFs is temporarily stored in SHTs 3 and 4 (Figure 1) until dewatering resumes on Monday morning. Circulation pumps are used to keep the sludge suspended in SHTs 3 and 4.

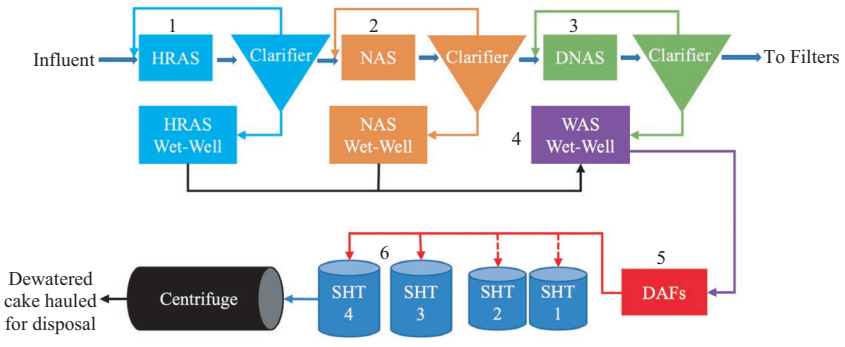
The dewatered biosolids produced from the centrifuge exhibits exceptionally high odor emission. Typical odor-causing compounds emitted from biosolids include hydrogen sulfide (H_2S) and volatile organic sulfur compounds (VOSCs) such as methanethiol (MT) and dimethyl sulfide (DMS) (Novak et al., 2006). It is widely accepted that these odorous compounds are mainly generated from anaerobically fermented organic materials (Higgins et al., 2006; Schlegelmilch et al., 2005). More specifically, H_2S and MT are the microbial degradation by-products of sulfur containing amino acids such as methionine and cysteine (Higgins et al., 2006), which have been shown to be present in protein extracted from activated sludge and anaerobically digested sludge (Dignac et al., 1998; Higgins et al., 2004; Morgan et al., 1991). Moreover, MT and DMS have also been found to be the products of methylation of H_2S and MT by anaerobic bacteria (Bak et al., 1992; Lomans et al., 1997). It should be noted that in the presence of sulfate, sulfite, or thiosulfate, sulfate reducing bacteria (SRB) are also able to convert these substrates to H_2S under anaerobic condition (McCarty & Smith, 1986).

Although there are various existing physical, chemical, and biological methods available for sulfurous odor control such as dry or wet adsorption, membrane separation, chemical precipitation, or even the replacement of high-shear centrifuge with low shear belt press (Erdal et al., 2008), these conventional approaches require significant capital, material, and/or chemical investment. In contrast, dosing air to improve the oxidation reduction potential (ORP) within the SHTs holds promise to provide a simple and economical solution for biosolid odor control. This is because previous studies have indicated that aeration can not only reduce the average hydrogen sulfide concentration in biogas of the anaerobic digesters (Krayzelova et al., 2014) but also reduce the odor emission from anaerobically digested biosolids (Winter et al., 2004). Although very little is known about the effect of SHT aeration on odor control of the biosolids produced from this WRRF without anaerobic digestion, it can be hypothesized that very similar mechanism can be expected in the anaerobic environment of the SHTs. Therefore, the objectives of this study were to verify this hypothesis and then develop a simple and cost-effective means for biosolids odor control. It is our anticipation that the outcome from this case study will provide engineering guidance for wastewater utilities with budget constraints and similar odor control needs.

MATERIALS AND METHODS

ORP profiles measurement

Two handheld ORP meters (MTC 101 probe, Hach, Loveland, Colorado) were used to measure replicate ORP



Information:

1) SHTs 1 and 2 are spare tanks equipped with pump circulation and aeration apparatus;
2) SHTs 3 and 4 are routinely used tanks equipped with pump circulation but without aeration apparatus.

FIGURE 1 Flow chart of the sludge handling system at the studied WRRF

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values throughout this WRRF's solids handling stream as shown in Figure 1. More specifically, two 3 L sludge samples were collected from locations 1 to 6 in Figure 1. Two ORP probes were inserted into the samples separately and allowed to stabilize for 5 min before reading. To increase the confidence of ORP profiles measured in this study, two separate ORP profiles were measured in two different field events on 25 days apart.

Sludge combination study

Fresh HRAS, NAS, and DNAS sludges with total solids (TS) of ~0.5% (g g^{-1}) were collected from the WRRF and blended at bench scale according to the actual blending ratios used in the full-scale WRRF. To infer the contribution of individual or combinations of sludges on the odor emission from dewatered cake, all seven possible combinations of the three types of sludge were tested (Table 1). The seven groups of blended sludge were then thickened using a laboratory centrifuge (Sorvall Legend X1R; Fisher Scientific, Waltham, MA) under a centrifugal acceleration of 2940 *g* for 2 min to achieve TS of ~4%, which is similar to the value after the DAFs at the full scale. Finally, the thickened sludge was placed in 300 ml enclosed bottles for 64 h to mimic sludge storage in the SHTs and then dewatered to approximately 20% TS using the laboratory-scale dewatering apparatus described below and used in a previous study (Zhang et al., 2020). TS and volatile solids (VS) contents of seven groups of dewatered biosolids were provided in Table 1. The seven groups of dewatered biosolids were then placed in 300 ml glass jars to perform odor emission analysis.

Bench-scale SHT aeration

Thickened sludge with TS of ~4% was collected from the end of the DAF system in Figure 1 and then divided into three groups: (1) Control: raw sample with neither mechanical mixed nor aeration; (2) Mechanical mixed without aeration; and (3) Mechanical mixed with aeration. Sludge samples with volume of 1 L were added into 1.5 L glass bottles. Mechanical mixing was provided with a magnetic stirring bar rotating at 150 rpm in order to mimic the pump circulation system in the full-scale SHTs. Aeration was provided by an air stone suspended in the middle height of the sludge bed with an air flow rate set at 2.7 L min^{-1}. All bottles were sealed with parafilm. A 1 cm diameter hole was punched on the parafilm to simulate the inadequate airtightness status of full-scale SHTs. All sludge groups were prepared in duplicate and held for 64 h with pH and ORP measured daily. After 64-h holding time, all groups were dewatered and then placed in glass jars for odor emission analysis.

Full-scale SHT aeration

Effect of aeration in the SHT on odor emissions from dewatered cake was also confirmed in a full-scale test. The studied WRRF has four SHTs, numbered SHTs 1, 2, 3, and 4 (Figure 1). Normally, only SHTs 3 and 4, which are larger in size and do not have aeration equipment, are used to hold thickened sludge prior to dewatering. In contrast, SHTs 1 and 2 (Volume ~1.2×10^5 L in each) are equipped with aeration diffuser grids and used for backup. In this study, continuous aeration at an air flow rate of 3540 L min^{-1} was provided to SHTs 1 and 2 using diffuser grids across the entire bottom of these two tanks for 65 h over the weekend until the start of the dewatering process on Monday. ORP values were measured for both aerated and non-aerated sludge after 65 h of holding in the full-scale SHTs. The sludge samples were then dewatered at full scale and divided into three groups depending on the aeration and autoclave treatment applied, that is, (1) mechanically mixed only; (2) mechanically mixed +aeration; and (3) mechanically mixed + aeration + autoclave. All the groups were prepared in duplicate. The sludge samples were autoclaved in capped jars at 120°C for 15 min. The purpose of autoclaving the sludge samples was to eliminate all biological activity to determine whether the odor was

TABLE 1 Seven combinations of the three types of thickened sludges

Group	HRAS:NAS:DNAS (V:V:V)	Dewatered cake TS (g g^{-1})	Dewatered cake VS (g g^{-1} TS)
HRAS	1:0:0	18%	82%
NAS	0:1:0	19%	77%
DNAS	0:0:1	20%	68%
HRAS + NAS	95:4:0	19%	83%
HRAS + DNAS	95:0:1	16%	82%
NAS + DNAS	0:4:1	20%	70%
3 in one	95:4:1	16%	82%

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controlled through biological activity inhibition or physical air stripping during the SHT aeration. Finally, the odor emission analysis was conducted for all three groups of sludges.

Dewatering at bench scale

A bench-scale dewatering system was established to mimic the mechanism of dewatering processes in full-scale application. Sludge samples (300 ml) were mixed with 80 ml polymer composed of 60% cationicity and 41% active polyacrylamide (Clarifloc, C-6226, Polydyne Inc, Riceboro, GA) and then sheared in a commercial blender (Waring MX1200XTXP; Stamford, CT) under controlled shear intensity quantified by $G \cdot t$ at 10^5 in which G is the mean velocity gradient and t is the time taken for sludge to be exposed to shear force. This shear intensity was calculated based on the information of the full-scale centrifuge used at the WRRF. The mixture was then preliminarily dewatered using a lab centrifuge at 5762 *g* for 15 min, followed by further mechanical compression to produce dewatered cake using a piston press under controlled pressure of 413 kPa and for 30 min. More detailed information about the dewatering apparatus can be found in our previous study (Zhang et al., 2020).

Odor emission analysis

Serum jars that were 300 ml in volume were loaded with ~20 g dewatered cake samples and then purged with nitrogen gas to remove headspace oxygen. The jars were sealed with rubber stoppers and incubated at the room temperature. Simultaneously, a 1 L air bag (Tedlar, Complex Plastics, Indiana UAS) was connected to normalize the air pressure inside of each jar. The concentrations of odorous compounds (H_2S, MT, and DMS) in the headspace of jars were monitored on a daily basis to obtain the odor emission profiles along with the sludge storage time.

Olfactometry analysis

Biosolid samples were sent to the Penn State University Olfactometry Lab to quantify the odor detection threshold (DT) using methods described elsewhere (Quality, 2003; St. Croix Sensory, 2003). The DT represents the lowest odor concentration that can be sensed by a panel of odor assessors as being different from odor-free air. For example, an odor DT of 1000 indicates that an odor can be sensed if diluted with odor-free air at a ratio of 1/1000. This is the metric mostly often used to define the odor concentration in environmental odor studies (WRF, 2019).

Chemical analysis

The contents of odor compounds such as H_2S, MT, and DMS in the headspace of jars were measured using a gas chromatograph (GC) equipped with a flame ionization detector (GC-2014, Shimadzu, Columbia, MD) and a 60 m long capillary column with 0.53 mm diameter (Agilent, DB-1, Santa Clara, California). Headspace gas volume of 40 ml was collected using a 50 ml gas-tight syringe (Hamilton, Reno, Nevada) on a daily basis and manually injected into the GC. Standard gases (Airgas, Inc., Radnor, Pennsylvania) were used to develop the calibration curves for these odorous compounds.

Statistical analysis

The values of ORP, pH, and concentrations of H_2S, MT, and DMS are presented as the sample mean with standard errors shown in the error bars.

RESULTS

Abnormally high odor emission from WRRF biosolids

According to the biosolids odor assessment results shown in Figure 2, the odor DT of lime-amended biosolids from the studied WRRF was reported as 2512, which was almost three times as odorous as lime-stabilized biosolids from WSSC Water's another WRRF (DT = 864) (Figure 2). The biosolids odor DT from the studied WRRF was also compared to previously published biosolids odor DT values by Water Research Foundation (WRF, 2019). In that report, biosolids samples taken from 16 WRRFs were reported to have odor DTs ranging from 126 to 3236 with a mean DT approximately 1133 (Figure 2). Only one of the 16 samples had an odor DT value greater than that of the biosolids from the WRRF investigated in this study (Figure 2). Based on this comparison, it is clear that the biosolids from this studied WRRF exhibited an abnormally high-odor DT value.

ORP profiles

Replicate ORP profiles in Figure 3 were measured at the studied WRRF during two separate field sampling events and showed very similar trends. The ORP dropped from 110 mV, 150 mV, and 120 mV in the HRAS, NAS and DNAS, respectively, to approximately −100 mV in the waste activated sludge (WAS) wet well once they were blended. Although the ORP remained unchanged in the DAF effluent

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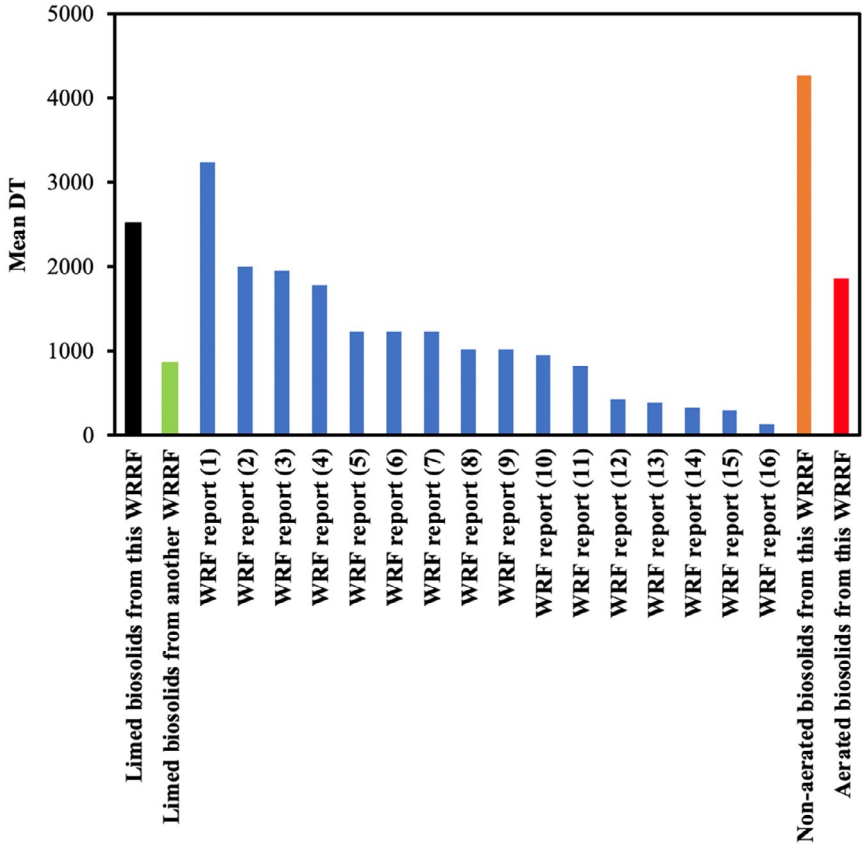


FIGURE 2 A comparison of limed biosolids DT values from this WRRF under current investigation and another WRRF administrated by WSSC Water, as well as the DT values of biosolids collected from 16 WRRFs as reported in WRF 2019 report. The biosolids DT values from this studied WRRF full-scale trial with and without SHT aeration are also shown

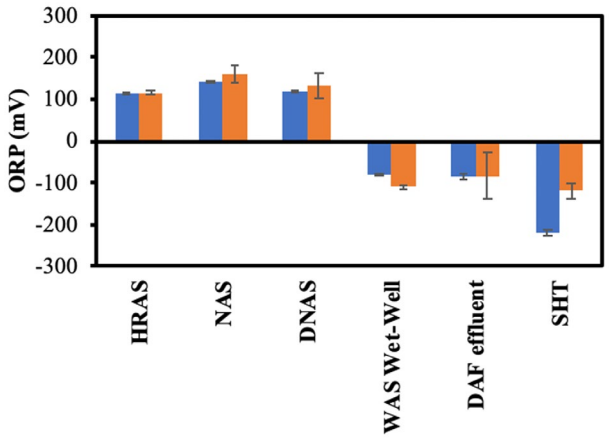


FIGURE 3 ORP profiles measured in two separate field trips sampling events at the studied WRRF

(approximately −90 mV), it drastically dropped again to approximately −170 mV in the SHTs, which created ideal anaerobic conditions for the formation of odorous compounds such as H_2S, MT, and DMS. During the sampling events, the odor seemed to be greatest around the SHTs.

Source of the odor formation

As shown in Figure 4, a matrix of seven sludge combinations were tested to evaluate the contribution of HRAS, NAS, and DNAS to odor emission from dewatered cake (Table 1). Whenever HRAS was blended in, substantial levels of H_2S, MT, and DMS were generated from the dewatered cake. This can be possibly explained by the high protein fraction typically contained in the HRAS. A similar observation was reported previously (Hentz et al., 2000), which found that storing primary and waste activated sludges in separate sludge holding tanks can achieve 50–70% less odor from dewatered biosolids than storing both sludges together. In comparison, only negligible levels of H_2S, MT, and DMS were formed in NAS, DNAS, and NAS + DNAS. Hence, NAS and DNAS appeared to have very little to do with the source of odor emission.

Effect of lab-scale SHT aeration

Aeration of thickened solids during the 64-h holding time was tested at bench scale as a potential odor mitigation strategy. As shown in Figure 5a, ORP in the aeration group increased

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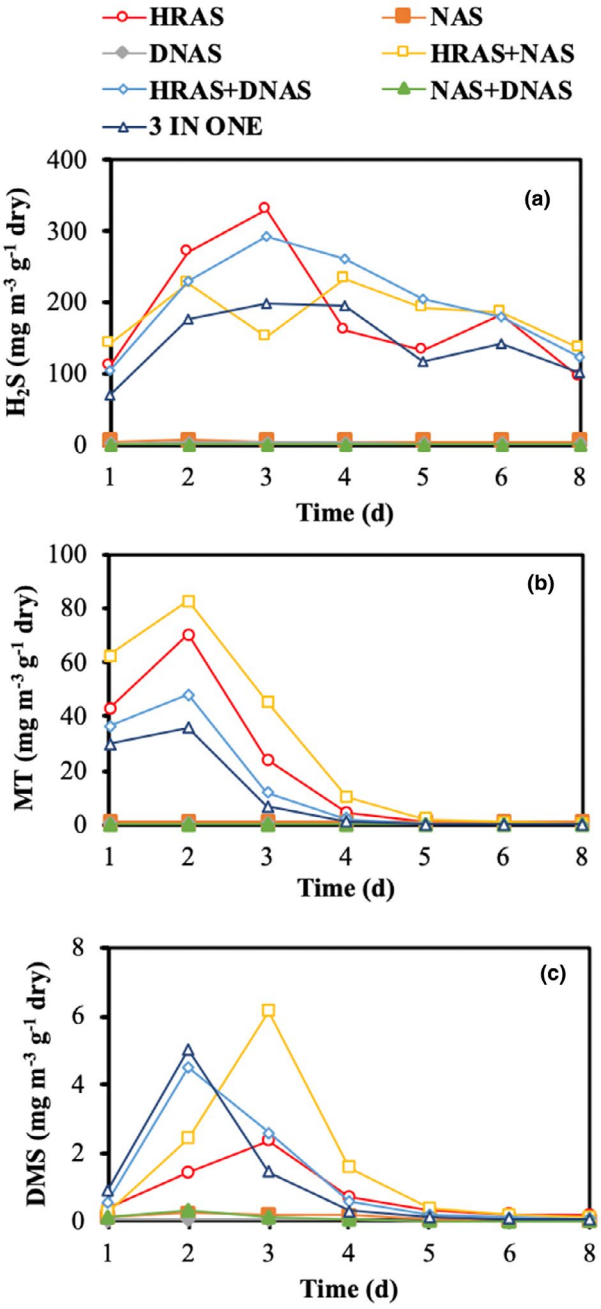


FIGURE 4 Profiles of (a) H_2S, (b) MT, and (c) DMS emission from dewatered cake during over time for the seven combinations described in Table 1

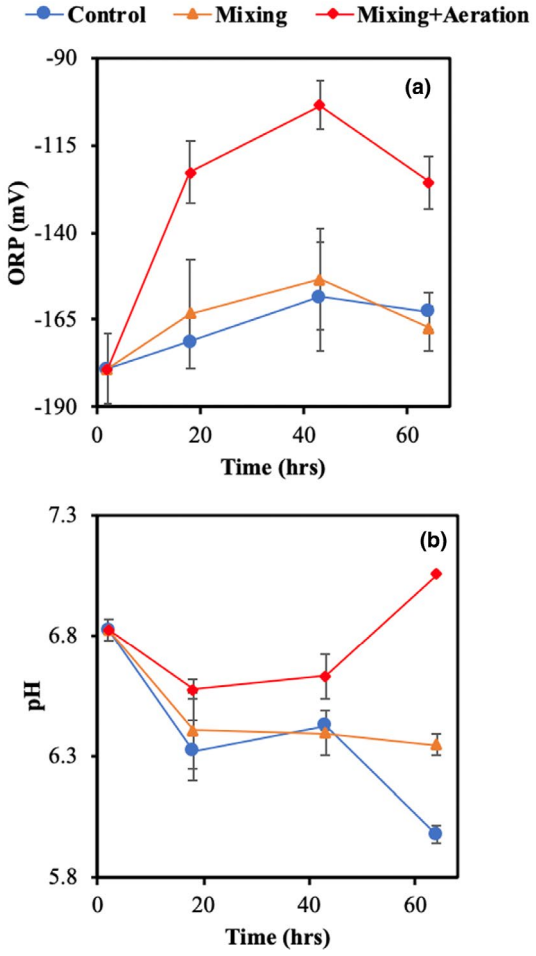


FIGURE 5 Profiles of (a) ORP and (b) pH measured during the 64-h holding time in the first round of bench-scale experiment mimicking the SHT environment

gradually from −179 mV to −120 mV with continuous aeration over the simulated 64-h storage time (i.e., became less anaerobic). The peak ORP was as high as −103 mV, which is approaching anoxic. In contrast, ORP values in the control group and mechanical mixed groups without aeration remained anaerobic at levels below −159 mV throughout the experiment. This indicates the inability of mechanical mixing, mimicking the pump circulation in full scale, to entrain enough air to increase the sludge ORP. Meanwhile, pH of the aeration group increased from 6.83 to 7.06 at the end of the holding time (Figure 5b). However, the pH in the other groups dropped from 6.83 to 5.98 and 6.35, respectively (Figure 5b). The plausible explanation is that anaerobic fermentation occurring in the mechanically mixed and control groups might have produced acids while the oxidization of organics in the aeration group produced alkalinity.

The three groups of sludges were dewatered after the 64-h holding time, and the dewatered cakes were stored in glass jars to perform the odor emission analysis. The odor compounds were monitored until the concentrations reached low levels and/or stabilized, which is beyond the typical 1- to 3-day dewatered cake storage time. The dewatered cake from the aerated group exhibited the lowest peak levels of H_2S, MT, and DMS throughout the 17-day monitoring period compared to the mechanical mixed group and the control group (Figure 6a,b). The most striking result was that the DMS emission peak for the cake that was aerated appeared the latest on day 8 and had the lowest peak DMS concentration (13.86 mg m^{-3} g^{-1}), which was approximately 30%

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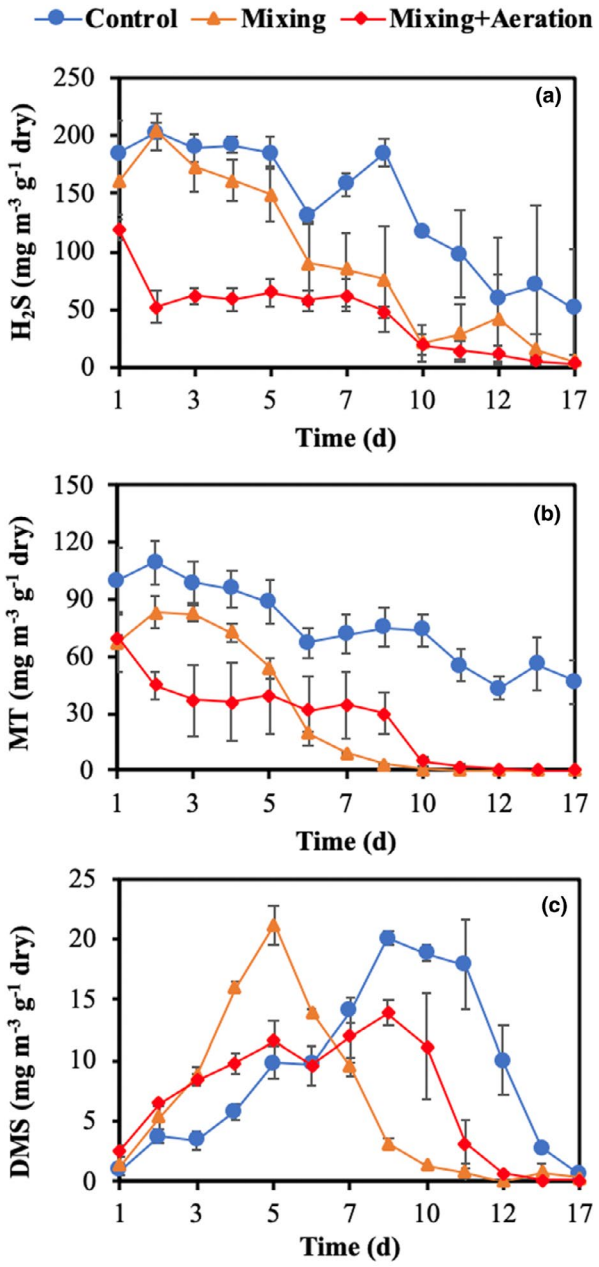


FIGURE 6 Profiles of (a) H_2S, (b) MT, and (c) DMS measured from dewatered cake over time after the three groups of thickened sludge were subjected to mechanical mixing and/or aeration at bench-scale. Results shown are for the first round of bench-scale testing and were consistent with the second round

lower than those of the other two groups (Figure 6c). This bench-scale experiment demonstrated that aeration during the thickened sludge holding time can reduce sulfurous odor emission from the dewatered cake.

To increase confidence in the results obtained from this first round of the bench-scale SHT aeration experiment prior to proceeding with a full-scale test, a second round of the bench-scale aeration testing was performed. The pH and ORP profiles from the second round of testing are shown in Figure S1. The ORP measured in the aeration groups increased gradually from −170 mV to −120 mV as a result of continuous aeration over the 64-h sludge holding time (Figure S1a), which was similar to the first-round experiment as shown in Figure 5a. The ORP in the control and mechanical mixed groups without aeration remained at a low level throughout the holding time, as well (Figure S1a). Meanwhile, pH in these three groups in Figure S1b also showed a similar trend as that in the first round of testing as shown in Figure 5b, that is, the pH of the aeration group increased from 7.37 to 7.55 at the end of the sludge holding time while the pH in the other groups dropped from 7.37 to 7.12 and 7.05, respectively.

The same dewatering and odor analysis procedures that were used in the first round were repeated for the second round of testing. As shown in Figure S2, the aerated sludge cake still had the lowest concentrations of odorous compounds in terms of H_2S, MT, and DMS throughout the 23-day monitoring period compared to the mechanical mixed and the control groups without aeration during the same duration (Figure S2a-c). Similarly, the cake from the control group had the highest H_2S, MT, and DMS peak levels, followed by the cake from the mechanically mixed group (Figure S2a-c). Based on similar data obtained in Figure 6 and Figure S2, it can be concluded with confidence that aeration during the sludge holding time is capable of reducing odorous compounds generated from the dewatered cake.

Preliminary full-scale trial of the SHT aeration strategy

After confirming favorable results in the two bench-scale SHT aeration tests, full-scale testing of aeration in SHTs 1 and 2 as an odor mitigation strategy was conducted. ORP was measured in the SHTs for both the aerated (SHTs 1 and 2) and non-aerated (SHTs 3 and 4) thickened sludges after 64 h of holding time. The ORP value was −162 mV in the aerated sludge after ~64 h of aeration, which was only about 18 mV higher than the ORP of the non-aerated sludge (−180 mV; data not shown). This ORP improvement in the full-scale aerated sludge was not as high as that in the bench-scale experiments (Figure 5a and Figure S1a). It is suspected that the aeration time for the samples collected at full-scale was below the targeted 64 h due to the tank filling schedule for the solids operation during that timeframe. Additional full-scale testing is planned.

Dewatered cake samples with or without aeration pretreatment in the full-scale holding tanks were collected after full-scale centrifugal dewatering but before lime addition. The odor emission profiles were obtained for three groups of dewatered samples. Consistent with bench-scale results, the full-scale aerated sludge cake had the lowest concentrations

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of H_2S emission throughout the 14-day monitoring period compared to the mechanical mixed cake during the same duration through pump circulation (Figure 7a,b). However, there was not much difference between the two groups in terms of MT concentration (Figure 7b). Moreover, only minor levels of DMS were produced (Figure 7c), indicating that the major odorous compounds emitted from the full-scale dewatered cake were H_2S and MT. In view of this experimental result inconsistency which probably can be attributed to the insufficient aeration, the DT values of dewatered cakes with and without aeration were measured and shown in Figure 2. As can be seen, the aerated sludge cake (DT = 1862) was approximately 56% less odorous than the non-aerated sludge cake (DT = 4266) (Figure 2), suggesting that aeration in the full-scale SHTs indeed mitigated the odor emission from dewatered biosolids.

It was also of interest to identify the mechanism causing the reduction in odor emission from the dewatered cake. It is our hypothesis that the aeration in the SHTs can inhibit the activity of odor-producing microorganisms during the holding time, and further affect the odor emitted from the dewatered cake. However, a previous study also indicated that the aeration of wastewater can reduce the H_2S emission in the wet wells of lifting stations by stripping off dissolved H_2S to the air (Tomar et al., 1995). Therefore, another plausible explanation is that the aeration in the SHTs can strip off the dissolved odorous compounds to the air, leading to lower concentrations of dissolved odorous compounds in the dewatered cake going through the dewatering process and thus resulting in even less volatilization after dewatering. The autoclave process designed in this experiment was to verify whether the physical volatilization of the odorous compounds was a primary contributor to the odor reduction by eliminating the biological processes occurring in the dewatered cake. As shown in Figure 7, only negligible odorous compounds in terms of H_2S and MT were emitted from sludge cake autoclaved in capped jars. The sludge treated under the same conditions but without being autoclaved produced much higher H_2S and MT levels (Figure 7a,b), proving that it is actually the change in biological but not the physical process that has reduced odor emission during the holding time in this study.

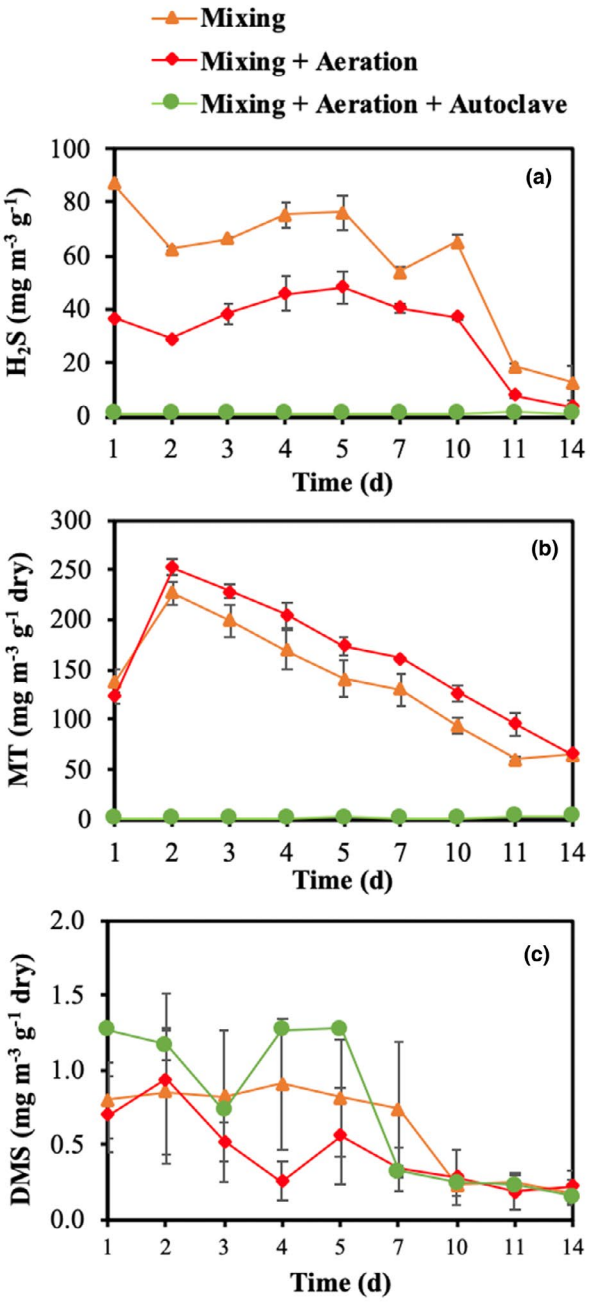


FIGURE 7 Profiles of (a) H_2S, (b) MT, and (c) DMS measured along with the sludge storage time for mixing only, mixing + aeration, and mixing + aeration + autoclave sludge samples taken from dewatering centrifuge in the full-scale study

DISCUSSION

Previous studies found that microaeration that increased the ORP above −100 mV can lead to the reduction of H_2S production as a result of the sulfate reduction activity inhibition (Ding et al., 2007). The average hydrogen sulfide removal in biogas of the anaerobic digesters can also be as high as 73% under microaeration condition in the anaerobic digester (Krayzelova et al., 2014). Likewise, previous work has also revealed that peak MT concentration under high-rate aeration (0.3 L kg^{-1} dry min^{-1}) during kitchen waste composting was four times lower than that under low rate aeration (0.1 L kg^{-1} dry min^{-1}) (Zhang et al., 2016). As a matter of fact, the effect of aeration on odor control has also been studied in the other fields, such as livestock manure slurry storage. For instance, 6.3 days of aerobic treatment of pig slurry achieved a reduction of the odor offensiveness rating from strongly offensive odor to faintly offensive odor

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(Burton et al., 1998). Although there are few published works related to the effect of SHT aeration in WRRFs, the results from this study revealed that even a slight increase of ORP from −180 mV to −162 mV in SHTs achieved through aeration might have compromised the activity of odor-producing bacteria in the dewatered cake.

The difference in terms of odorous compound emission from the sludge with and without being autoclaved suggests that it was the reduced anaerobic biological activity but not the air stripping during aeration that has primarily contributed to the reduction of odorous compound emission during the sludge holding time (Figure 7a,b). Therefore, it is highly possible that the ORP increase as a result of the aeration might have inhibited the activity of odor-producing bacteria in the dewatered sludge cake. Previous work indicated that the sulfate reduction predominated within the ORP range of −300 mV to −400 mV, while the ORP range of −50 mV to −150 mV as was achieved in SHT aeration in Figure 5a and Figure S1a was actually predominated by denitrification (Zhang et al., 2014). Likewise, the population of SRB was found to decrease by three to five orders of magnitude after 2 days of exposure to oxygen (Sass et al., 2002), and only a few species of SRB can escape oxygen toxicity (Krekeler et al., 1998). Thereby, it can be inferred that the aeration in the SHTs must have inhibited the activity of odor-producing bacteria and in turn controlled odor emission from later dewatered sludge cake.

PRACTICAL IMPLICATIONS

Both bench-scale and full-scale aeration experimental results in this study indicate that aeration in the SHTs during sludge holding time can effectively reduce the odor generation from the dewatered biosolids. Comparing to the other odor control methods such as membrane separation and chemical precipitation, the only cost of aeration is the air apparatus and energy, which is presumably economical but will need a techno economic assessment (TEA) to confirm. In addition, the potential adverse effect of SHT aeration should be also taken into consideration. For example, the stripped air from SHTs may need further purification to avoid air pollution. Also, some study reported that aeration can oxidize the sulfide in the sludge to elemental sulfur, which may deposit on the walls and pipes (Krayzelova et al., 2015).

Full-scale test in this study showed limited reduction in odors, likely due to insufficient airflow and/or duration in the tested holding tanks. Therefore, in the next phase of study, additional full-scale testing will be completed for SHTs 1 and 2 to see if a more controlled or longer aeration duration in the sludge holding tanks is more effective in reducing odors. An engineering consulting firm is also being retained by WSSC Water to evaluate and develop costs for several solids aeration alternatives for SHT 3 and 4 including aeration with the existing blower system, a new blower, and Venturi aspiration of recycle pumps.

CONCLUSIONS

The objectives, hypothesis, approaches, and results of this study were summarized in Table 2, and the following concluding remarks can be drawn from this study:

1. Abnormally high biosolids odor can be largely attributed to the storage of thickened sludge in the absence of oxygen for >48 h.
2. HRAS but not NAS and DNAS turned out to be the major contributor to biosolids odor emission.
3. Lifting ORP levels in SHTs through aeration should be considered as a cost-effective measure for biosolids odor control.
4. The SHT aeration is thought to have inhibited the activity of odor-producing bacteria and in turn reduced odor emission from dewatered biosolids.
5. Further full-scale validation and TEA are under way to confirm the cost-effectiveness of the approach.

TABLE 2 Summary of the study

Objective	Hypothesis	Experiment	Results
Identify cause of the odor	Low ORP is the major cause	ORP profiles were plotted throughout the entire treatment train	The lowest ORP spot is in the SHTs where ideal conditions for odor formation were created
Identify source of the odor	One of the three types of sludge is the main source	A full combination of all three types of sludges were tested for odor emission	Whenever HRAS was blended, the odor became prominent
Validate solution for odor control	Improvement of ORP by aeration may control the odor	Aeration was provided in both bench-scale and full-scale SHTs to verify its effectiveness	Bench-scale aeration test substantially increased ORP and thus effectively reduce odor, while full-scale test improved ORP to a certain extent and thus the odor control was not as effective as in the bench test

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AUTHOR CONTRIBUTIONS

Hao Luo: Data curation (equal); formal analysis (equal); methodology (equal); writing-original draft (equal). **Dian Zhang**: Conceptualization (equal); investigation (equal); methodology (equal). **Malcolm Taylor**: Investigation (equal); methodology (equal); project administration (equal); supervision (equal); writing-review & editing (equal). **Caroline Nguyen**: Methodology (equal); project administration (equal); supervision (equal), writing-review & editing (equal). **Zhi-Wu Wang**: Methodology (equal); project administration (equal); supervision (equal); writing-review & editing (equal).

ORCID

Dian Zhang https://orcid.org/0000-0001-5402-3898
Zhi-Wu Wang https://orcid.org/0000-0002-3496-1884

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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